1. What is software testing, and why is it important?

Definition: Software testing is the process of checking and verifying that a software application works properly according to its requirements and is free of bugs before it reaches the end user

Why it is important:

- It helps find bugs early, when they are usually easier to fix.
- It checks that the main features work properly.
- It can help find problems that may affect user data or security.

Question 2: Explain the difference between manual testing and automation testing.

Manual Testing: A testing process where a human tester executes test cases manually without using any automation tools or scripts to find defects and verify software behavior.

Automation Testing: A testing process where pre-scripted automation tools and code are used to execute test cases automatically, compare actual outcomes with expected outcomes, and report results.

Comparison: Manual Testing vs. Automation Testing

Parameter	Manual Testing	Automation Testing
Execution	Executed manually by a human QA engineer step by step.	Executed automatically using scripts and tools (e.g., Selenium, Playwright).
Speed & Efficiency	Slower and time-consuming, especially for repetitive flows.	Extremely fast; tests can run concurrently and 24/7 without intervention.
Initial Cost & Setup	Low initial cost; requires standard test design without tooling overhead.	High initial investment for tool licensing, test frameworks, and maintenance.
Human Error Risk	Prone to fatigue, oversight, and human error during long test cycles.	Highly accurate and consistent; executes exact instructions every run.
Programming Skills	No programming or coding knowledge required.	Requires programming language knowledge (e.g., Java, Python, JavaScript).

Ideal Use Cases	Exploratory testing, UI/UX evaluation, usability, and ad-hoc testing.	Regression testing, load/performance testing, and repetitive sanity checks.

Q3. What is the difference between functional and non-functional testing?

Functional testing : checks whether a feature does what it is supposed to do. For example, we check if a user can log in with the correct username and password, or add a product to the cart.

Non-functional testing : checks how well the application works. This can include speed, security, stability and how the website looks on different screen sizes. For example, checking whether a page loads in a reasonable time

Q4. Explain smoke testing, sanity testing, and regression testing.

Smoke testing: This is a quick check of the main features after receiving a new build. For example, checking whether the app opens and login works. If basic features fail, the build may not be ready for further testing.

Sanity testing: This is a quick check of a particular area after a small change or bug fix. For example, checking the login page again after a login bug has been fixed.

Regression testing: This checks that existing features still work after a change. For example, after updating checkout, we also check that login and the shopping cart still work.

Q5. What is the difference between bug severity and bug priority? Provide an example.

Bug severity means how badly the bug affects the application.
Bug priority means how soon the bug should be fixed.

Example 1: If the checkout page crashes and users cannot place an order, it has high severity and high priority because an important feature is not working.

Example 2: If the company name has a small spelling mistake on the home page, the issue may have low severity but high priority because it is visible to many users and should be corrected soon.